



LAB REPORT

Software Quality Assurance

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ROLL NUMBER:	Fall-23-BSCS-466
SUBJECT:	Software Quality Assurance
SECTION:	L(12)
SEMESTER:	6 th

Activity 1 — Decision Table Creation

Scenario: Online Shopping Discount System

Decision Table

Conditions / Actions	Rule 1	Rule 2	Rule 3	Rule 4
Premium Member	Y	Y	N	N
Purchase > Rs. 5000	Y	N	Y	N
Coupon Applied	Y	N	Y	N
20% Discount	Y	N	N	N
10% Discount	N	Y	Y	N
No Discount	N	N	N	Y

Generated Test Cases

TC ID	Premium Member	Purchase > 5000	Coupon Applied	Expected Result
DT-01	Yes	Yes	Yes	20% Discount
DT-02	Yes	No	No	10% Discount
DT-03	No	Yes	Yes	10% Discount
DT-04	No	No	No	No Discount

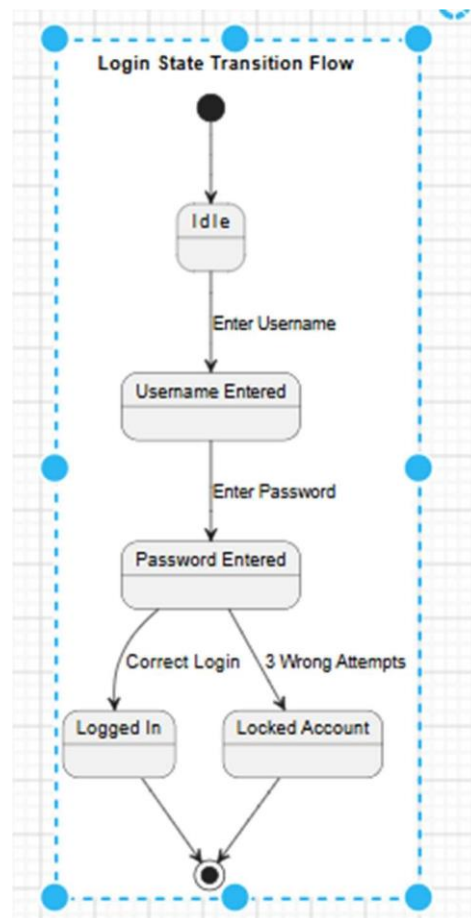
Activity 2 — State Transition on Testing

Scenario: University Portal Login

States

- Idle
- Username Entered
- Password Entered
- Logged In ☐ Locked Account

State Transition on Flow



Valid Transitions

Current State	Input	Next State
Idle	Enter Username	Username Entered
Username Entered	Enter Password	Password Entered
Password Entered	Correct Login	Logged In
Password Entered	3 Wrong A empts	Locked Account

Transition on Test Cases

TC ID	Current State	Input	Expected State
ST-01	Idle	Enter Username	Username Entered
ST-02	Username Entered	Enter Password	Password Entered
ST-03	Password Entered	Correct Login	Logged In
ST-04	Password Entered	3 Wrong A empts	Locked Account

Activity 3 — Specification on-Based Testing

Scenario: Student Registration on Form Requirements

- Name cannot be empty
- Email must contain @
- Password minimum 8 characters
- Age must be greater than 18

Test Conditions

1. Empty name validation
2. Invalid email validation
3. Password length validation
4. Age validation
5. Successful registration

Valid and Invalid Test Cases

TC ID	Input	Expected Result
SB-01	Empty Name	Error Message
SB-02	Email without @	Validation Error
SB-03	Password less than 8 chars	Registration on Rejected
SB-04	Age = 17	Registration on Failed
SB-05	All Valid Inputs	Registration on Successful

Activity 4 — Use Case Testing

Scenario: ATM Withdrawal System

Use Case Description

Item	Description
Use Case	Withdraw Cash
Actor	Customer
Preconditions	ATM card inserted
Postconditions	Cash dispensed

Main Flow

1. Insert ATM card

2. Enter PIN
 3. Select withdrawal op on
 4. Enter withdrawal amount
 5. Receive cash Alternate Flow
- Invalid PIN
 - Insufficient balance
 - ATM out of cash

Use Case Test Cases

TC ID	Scenario	Expected Result
UC-01	Valid withdrawal	Cash dispensed
UC-02	Invalid PIN	Error displayed
UC-03	Insufficient balance	Transaction on declined
UC-04	ATM out of cash	Service unavailable

Unsolved Exercises — Solved

Exercise 1 — Decision Table for Online Exam Submission System

Conditions ☐ Student Logged In

☐ Exam Time Active ☐ Internet Available

Actions

- Submission Successful
- Submission Failed

Decision Table

Conditions / Actions	R1	R2	R3	R4
Student Logged In	Y	Y	N	Y
Exam Time Active	Y	N	Y	Y
Internet Available	Y	Y	Y	N
Submission Successful	Y	N	N	N
Submission Failed	N	Y	Y	Y

Test Cases

TC ID	Conditions	Expected Result
EX1-01	Login + Time Active + Internet	Submission Successful
EX1-02	Login + Time Expired	Submission Failed
EX1-03	Not Logged In	Submission Failed
EX1-04	No Internet	Submission Failed

Exercise 2 — State Transition on Diagram for Traffic Light System

States

- Red
- Yellow
- Green

State Transitions

Red → Green

Green → Yellow

Yellow → Red

Test Cases

TC ID	Current State	Event	Next State
TL-01	Red	Timer Complete	Green
TL-02	Green	Timer Complete	Yellow
TL-03	Yellow	Timer Complete	Red

Exercise 3 — Specification on-Based Testing for E-commerce Checkout Form

Requirements

- Name cannot be empty
- Address required

- Card number must be 16 digits
- CVV must be 3 digits

Test Cases

TC ID	Input	Expected Result
EC-01	Empty Name	Valida on Error
EC-02	Empty Address	Checkout Failed
EC-03	Card Number less than 16 digits	Invalid Card Error
EC-04	CVV less than 3 digits	Invalid CVV Error
EC-05	All Valid Inputs	Order Placed Successfully

Exercise 4 — Use Case Testing for Food Delivery Mobile App

Use Case Description

Item	Description
Use Case	Order Food
Actor	Customer
Preconditions	User logged in
Postconditions	Order placed successfully

Main Flow

1. Open app
 2. Select restaurant
 3. Choose food items
 4. Add to cart
 5. Make payment
 6. Confirm order
- Alternate Flow
- Payment failed
 - Restaurant unavailable

- Item out of stock

Test Cases

TC ID	Scenario	Expected Result
FD-01	Successful order	Order confirmed
FD-02	Payment failed	Payment error shown
FD-03	Restaurant unavailable	Restaurant closed message
FD-04	Item out of stock	Item unavailable message